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Semester: 5th	Subject: Linux Operating System
Div: 1	Class: WTYBCA 2025/26

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UNIT 1: Introduction to Linux Operating System	
1. Display the Linux kernel version and system architecture.	
2. Identify the current shell and display its version.	
3. Switch to another shell (if available) and return back to Bash.	
4. Create a text file, a binary file, and list two special files	
5. Identify file types using appropriate commands.	
6. List contents of /bin, /etc, /home, and /var.	
7. Display the directory tree of your home directory.	
8. Check whether Linux is case-sensitive by creating two similar filenames.	
9. Navigate to /proc and observe any three files.	
10. Find the absolute path of your current working directory.	
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1. Display the present working directory.	
2. Create a directory named LinuxLab.	
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5. List all files including hidden files.	
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7. Remove an empty directory.	
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14. Create a file test.sh and display its permissions.	
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21. Display all logged-in users.	
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25. Clear the terminal screen..	
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1. Create a shell script to print “Welcome to Linux”.	
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16. Write a script to check whether a file exists	
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19. Write a script using case to implement a simple calculator.	
20. Write a script to check whether a number lies between 10 and 50.	
21. Write a script to check whether a file exists and is readable.	
22. Write a script to check whether a file exists.	
23. Write a script to compare two numbers.	
24. Write a script to compare two strings.	
25. Write a script to add two numbers using expr.	
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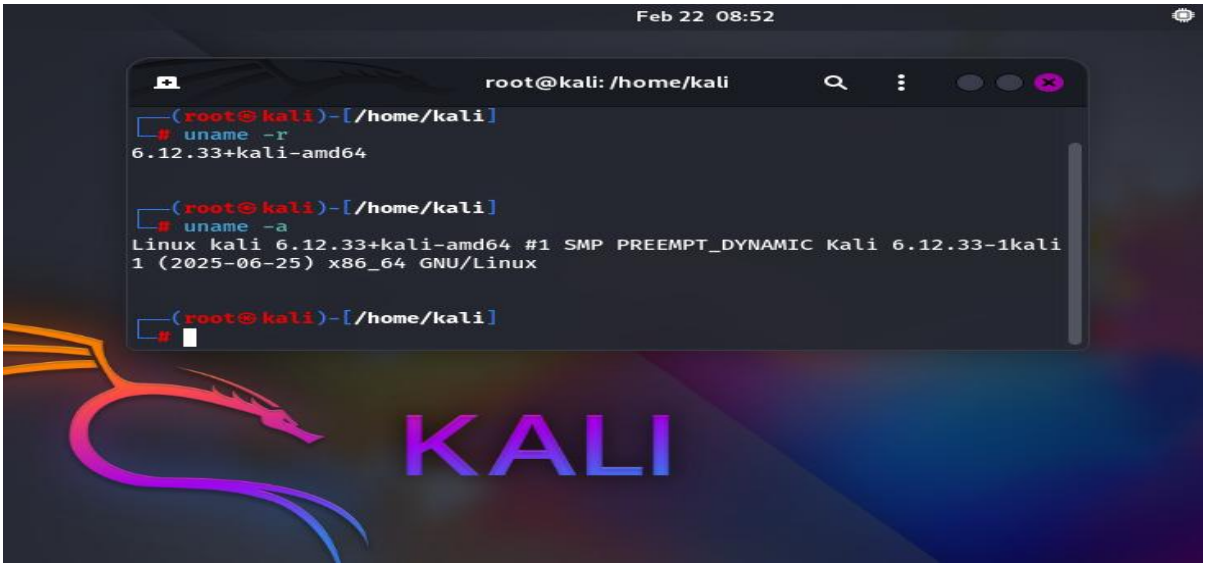
UNIT 4: Advanced Text Processing Tools	
1. Find all lines starting with the word “Linux” from a file.	
2. Find all lines ending with .txt.	
3. Find all lines containing exactly a 10-digit number.	
4. Find all lines containing only alphabets.	
5. Search for the word “error” in a file.	
6. Search lines containing digits using egrep.	
7. Search an exact string using fgrep.	
8. Display line numbers along with matched lines.	
9. Perform case-insensitive searching.	
10. Replace the word “Linux” with “UNIX” in a file.	
11. Delete the 3rd line from a file.	
12. Insert a line at the beginning of a file.	

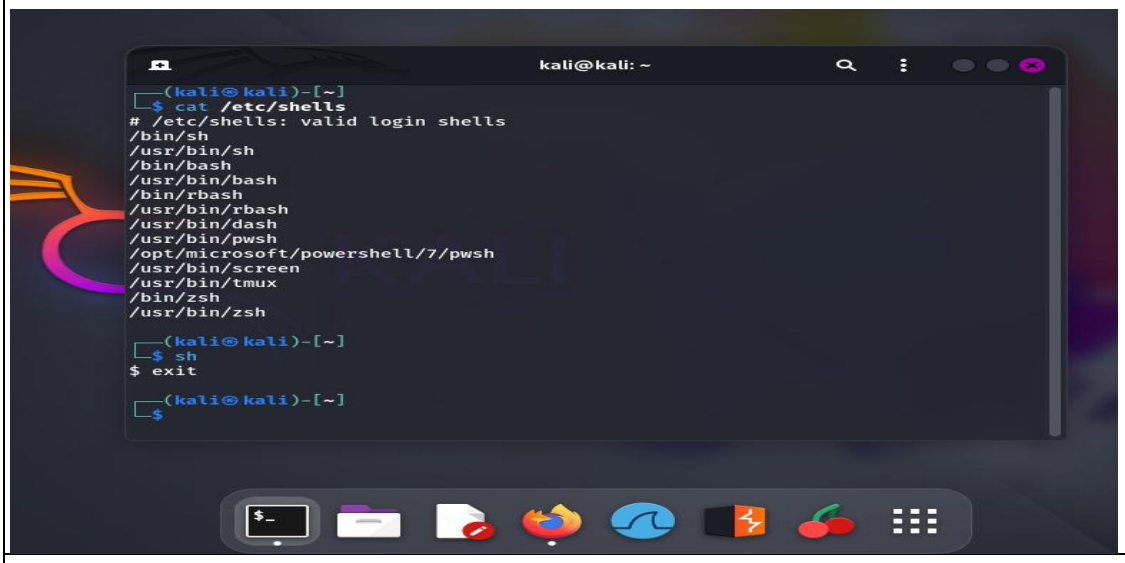
Unit-1

1. Display the Linux kernel version and system architecture.



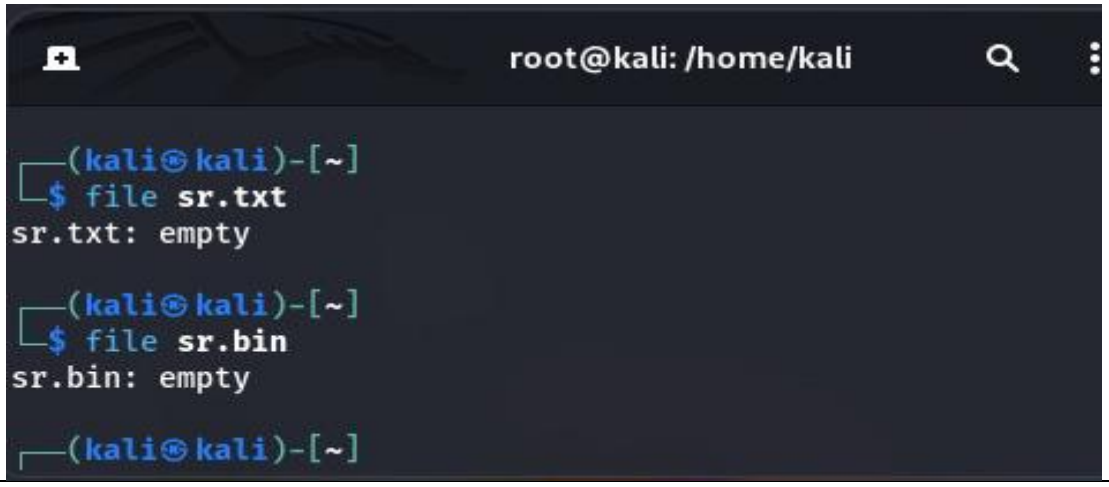
2. Identify the current shell and display its version.

3 Switch to another shell (if available) and return back to Bash.

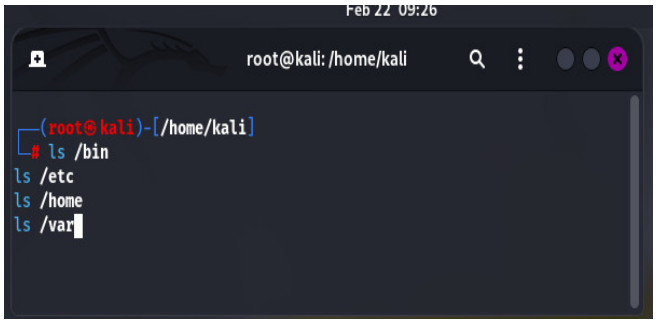


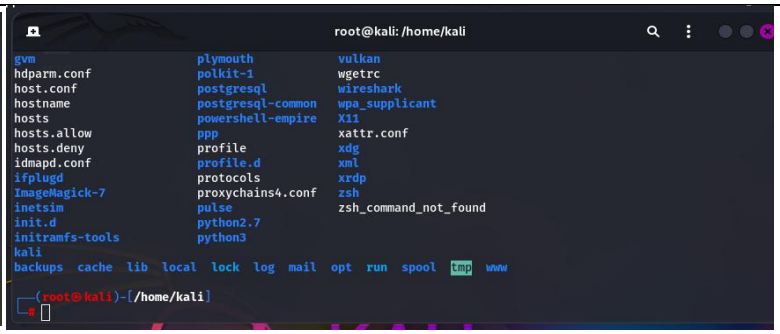
4 Create a text file, a binary file, and list two special files.

5 Identify file types using appropriate commands.



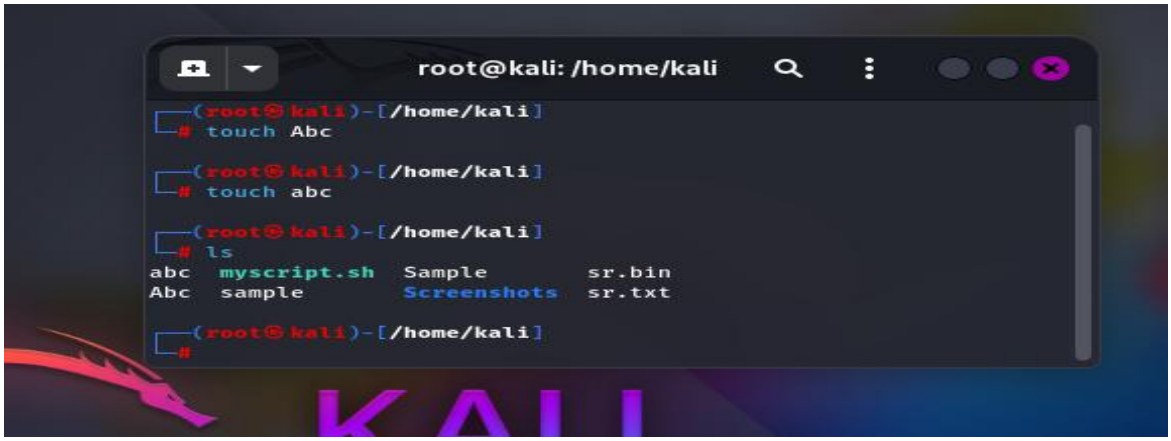
6 List contents of /bin, /etc, /home, and /var.



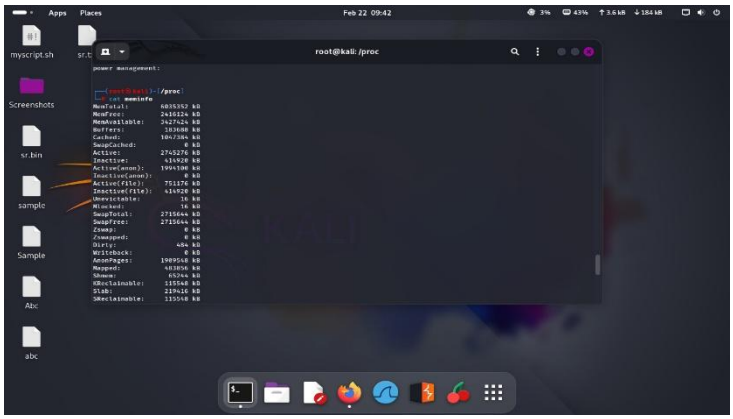


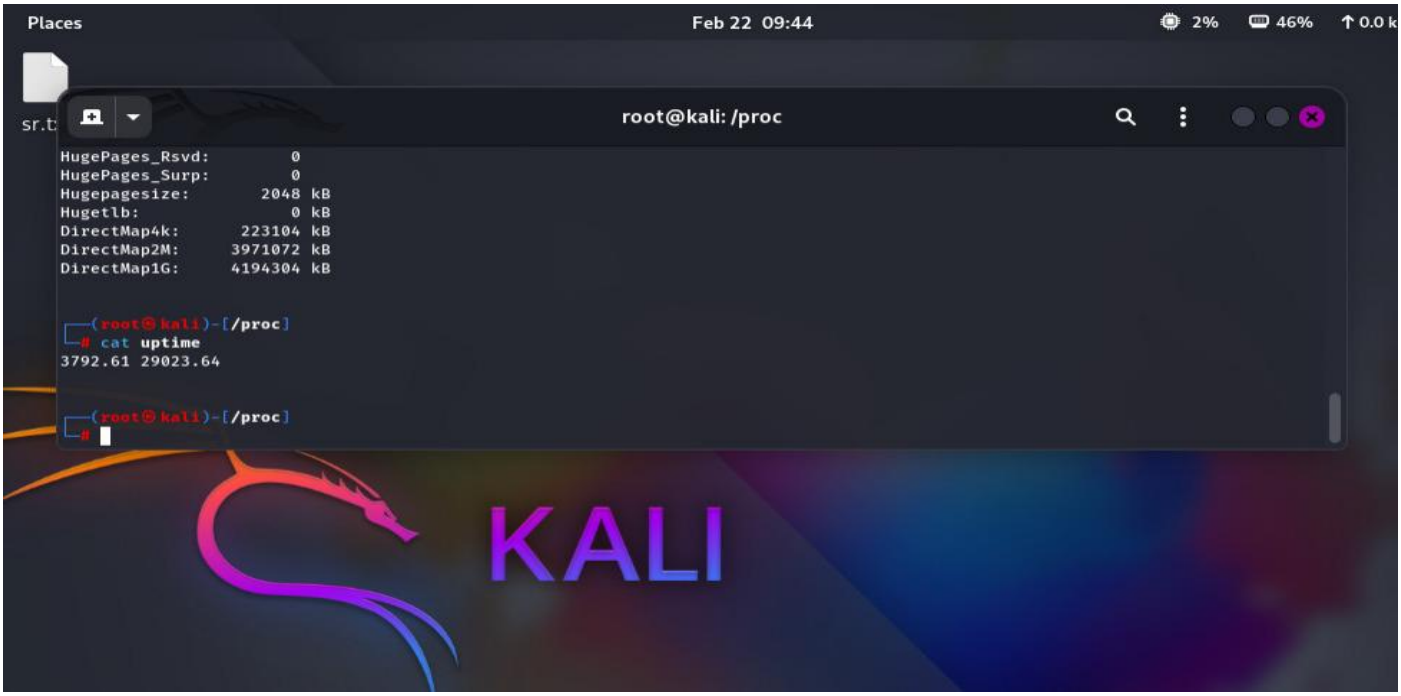
7 Display the directory tree of your home directory.

8. Check whether Linux is case-sensitive by creating two similar filenames.



9 Navigate to /proc and observe any three files.





10. Find the absolute path of your current working directory.

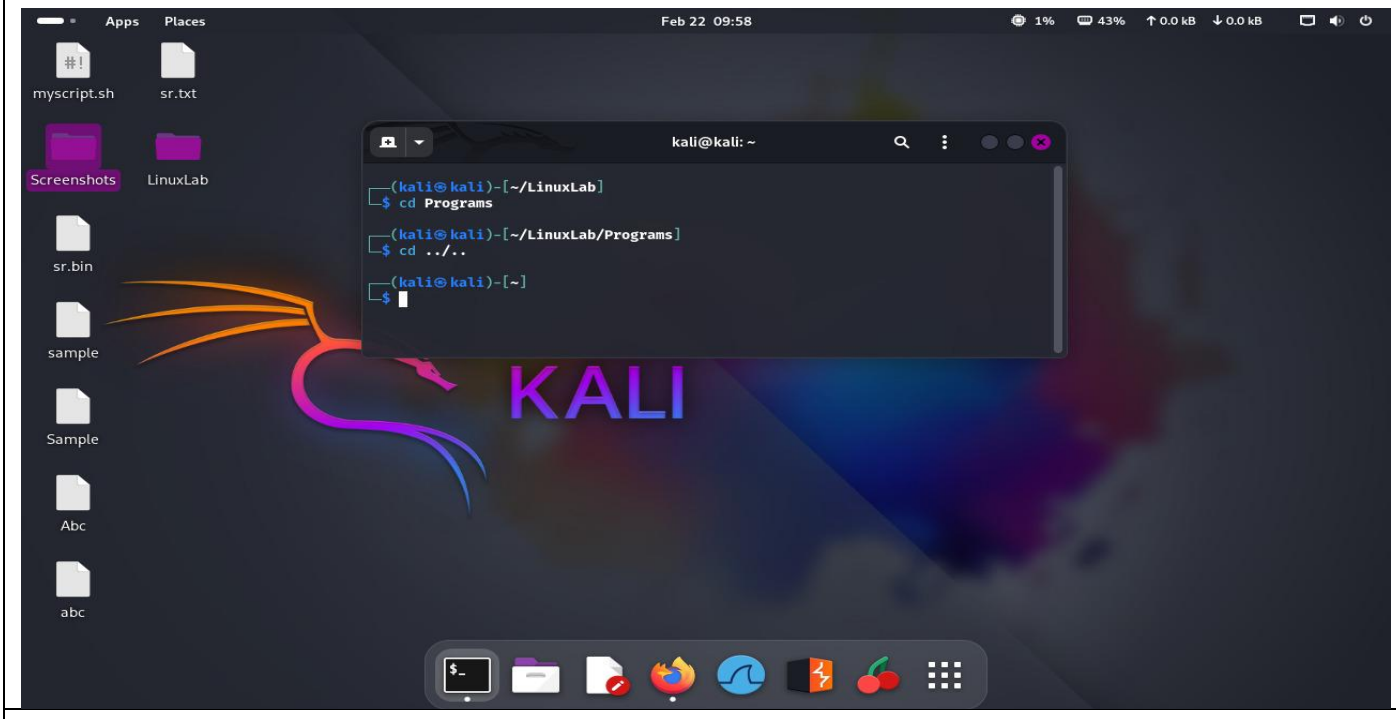
Unit -2

1 Display the present working directory.

2 Create a directory named LinuxLab.

3 Inside LinuxLab, create subdirectories Docs, Programs, and Output.

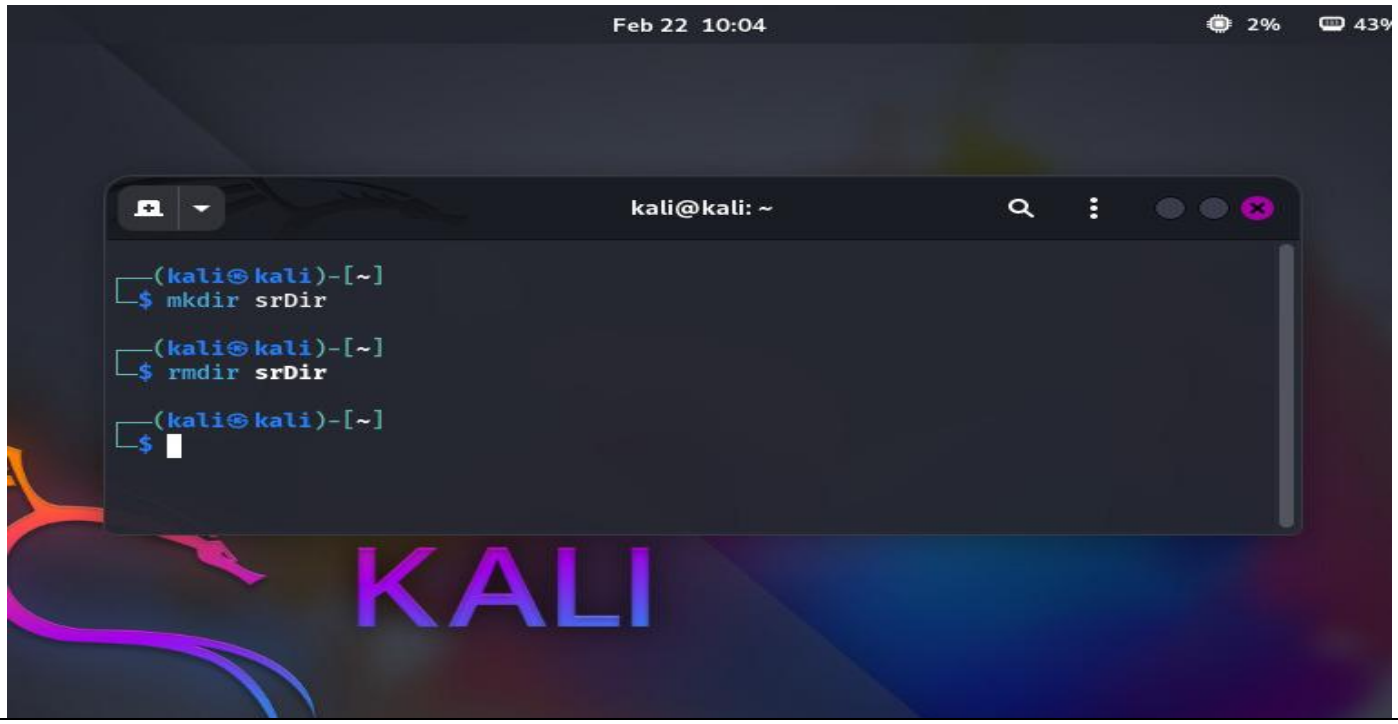
4. Navigate into Programs and return to the home directory using relative paths.



5. List all files including hidden files.

6 Display the directory structure of your home directory using tree.

7 Remove an empty directory.

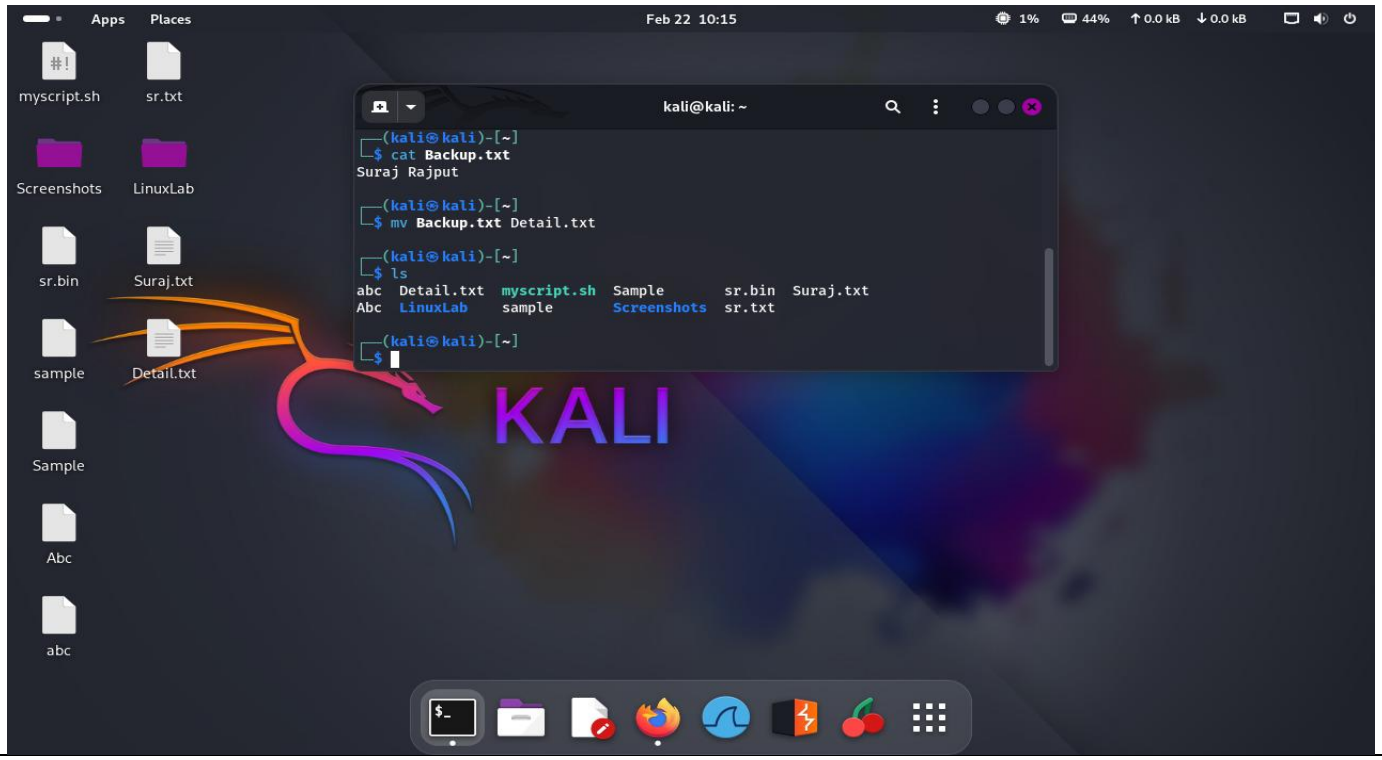


8 Create a file Suraj.txt and write your name and class in it.

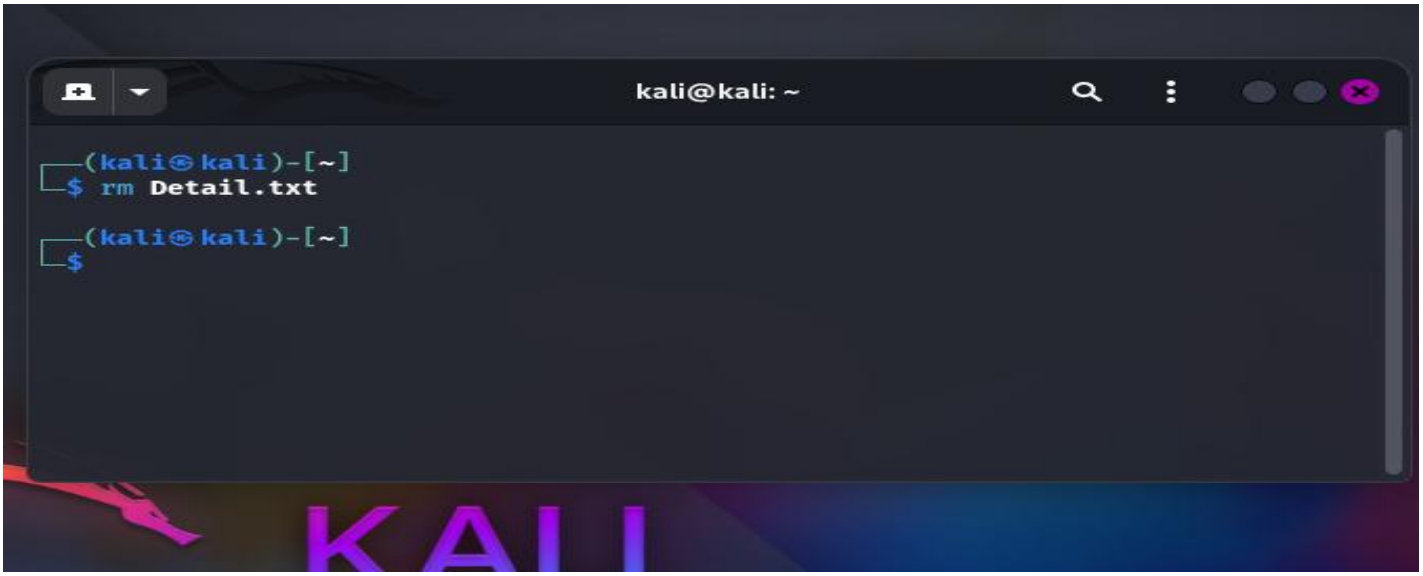
9 Display the contents of Suraj.txt.

10 Copy info.txt to backup.txt.

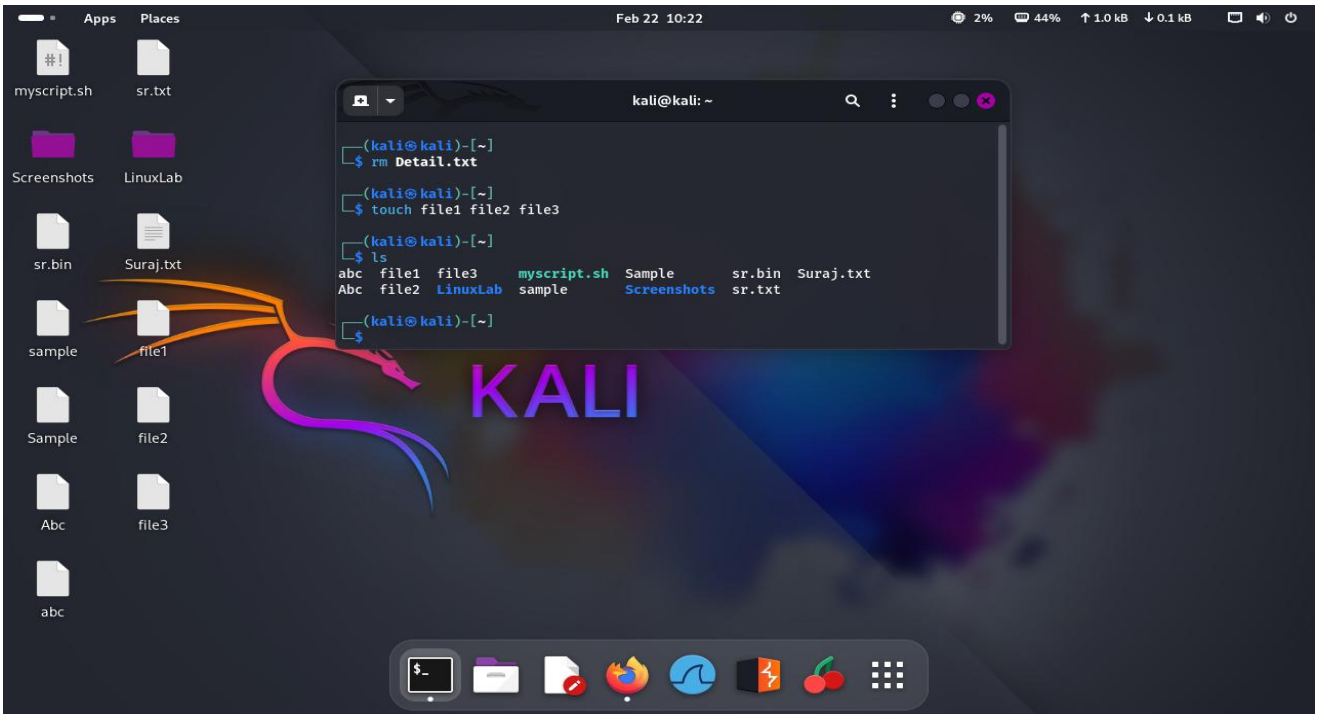
11 Rename backup.txt to details.txt



12 Delete details.txt.

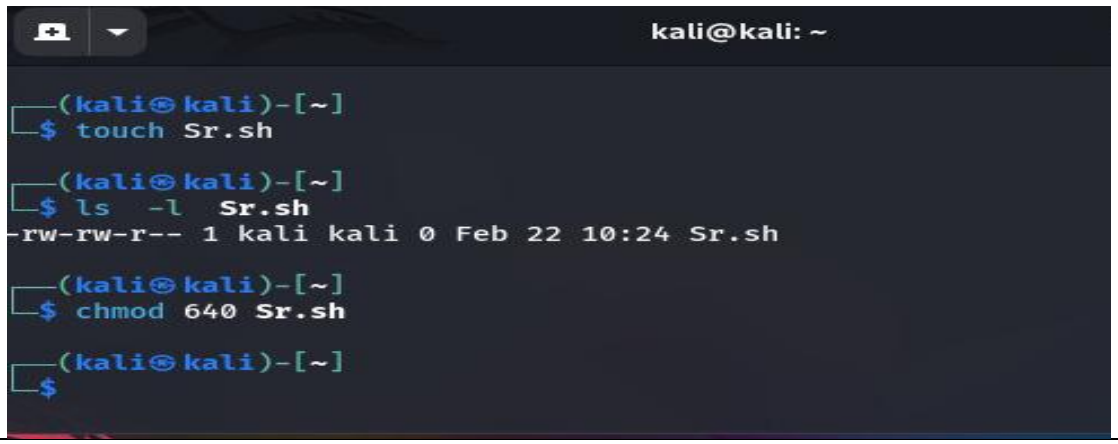


13 Create three files in one command using touch.

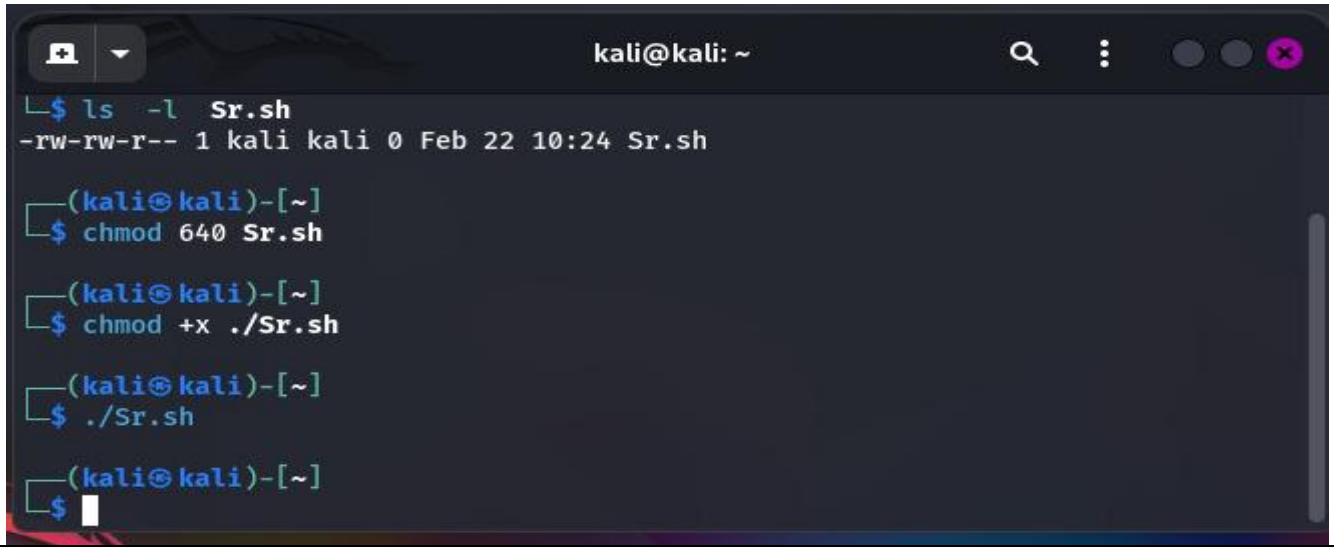


14 Create a file test.sh and display its permissions.

15 Change permissions so that owner has read-write, group has read, and others have no permission.



16 Make test.sh executable and run it.

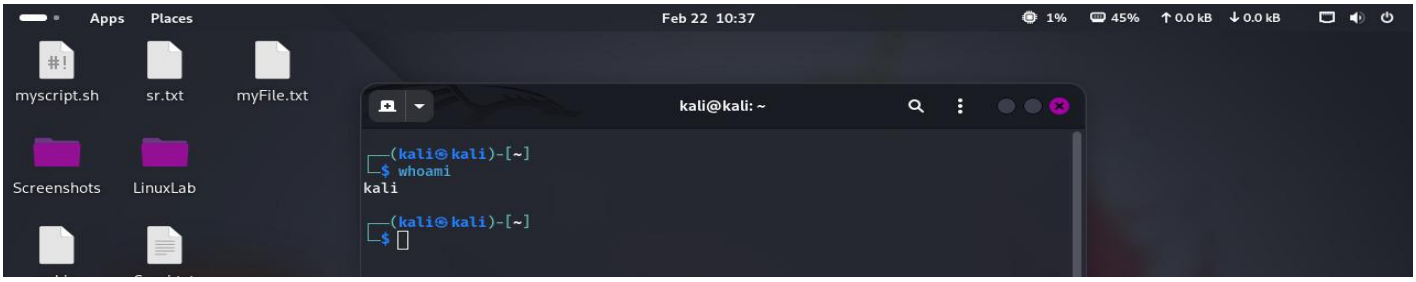


17 Change the group ownership of test.sh.

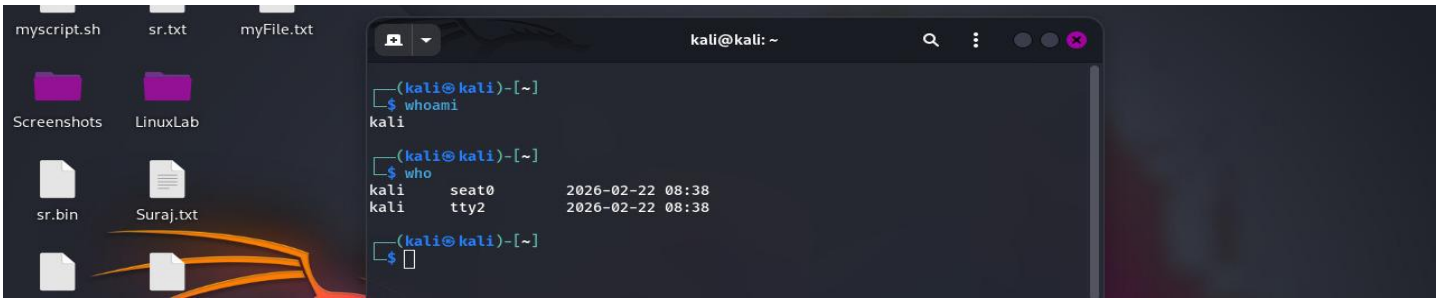
18 Display the current umask value

19 Create a new file and verify default permissions.

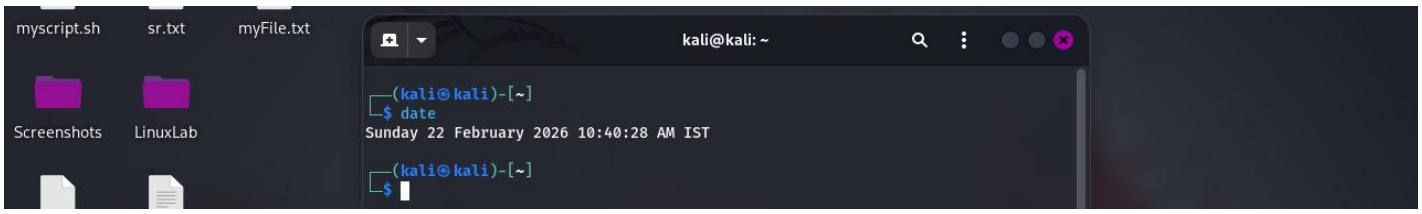
20 Display the current logged-in user.



21 Display all logged-in users.

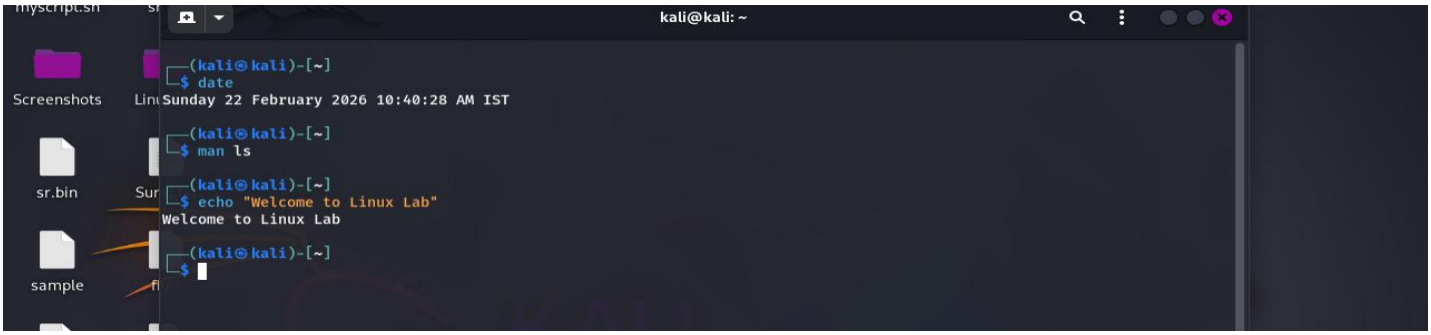


22 Display today's date and time.



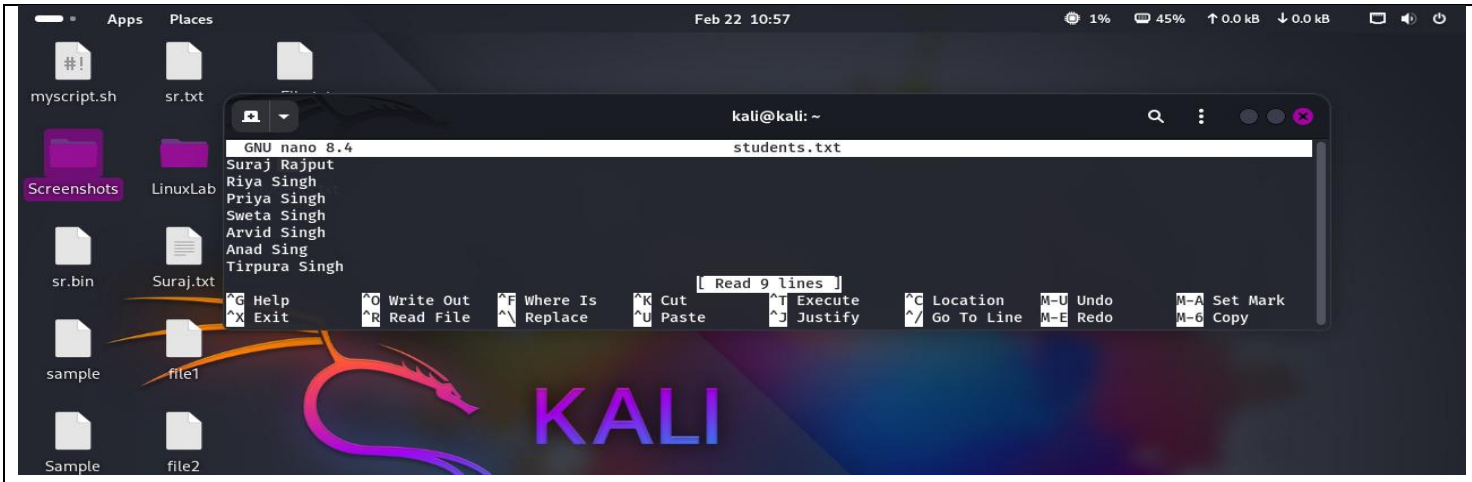
23 Display the manual page of ls.

24 Print the message “Welcome to Linux Lab”.

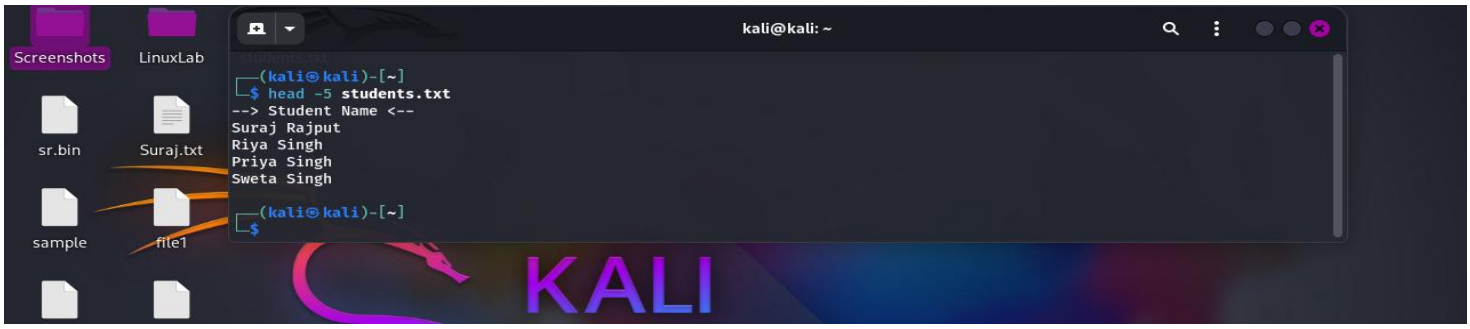


25 Clear the terminal screen.

26 Create a file students.txt with at least 10 student names.



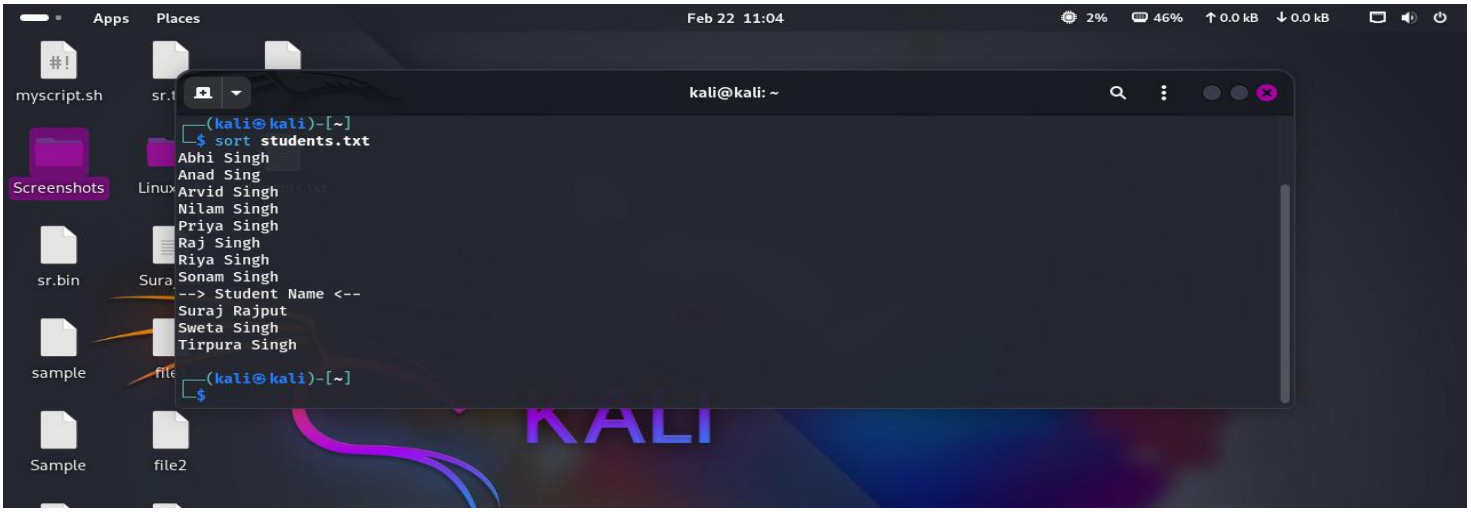
27 Display the first 5 lines of the file



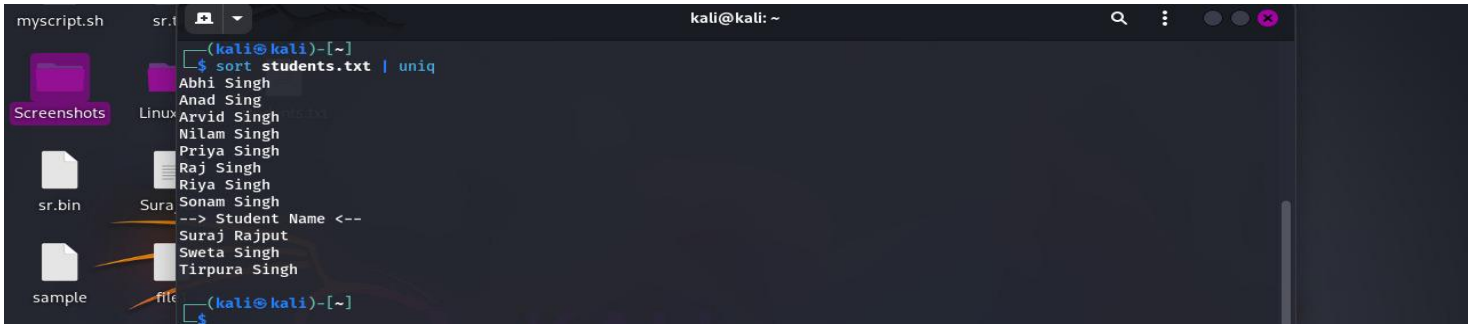
28 Display the last 3 lines of the file

29 Count the number of lines, words, and characters in the file.

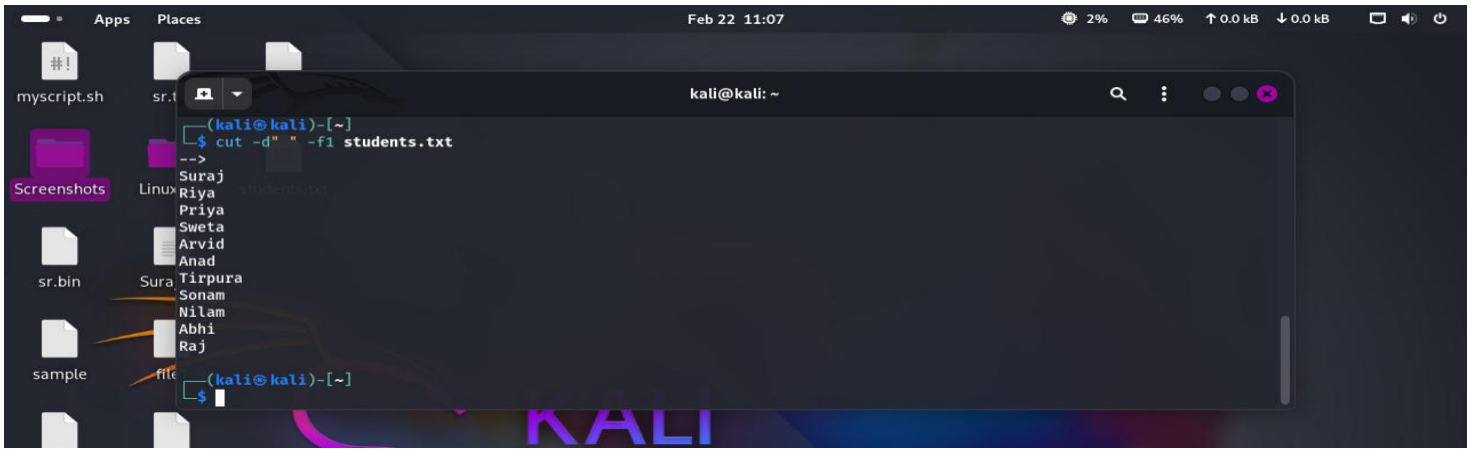
30 Sort the file alphabetically.



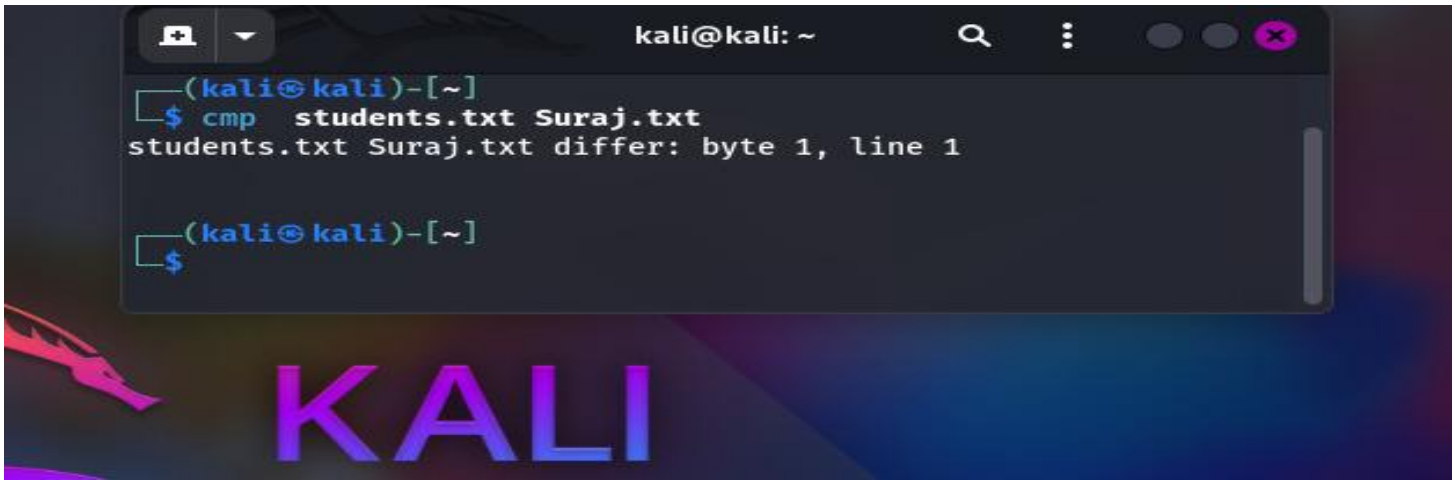
31 Remove duplicate lines from the file.



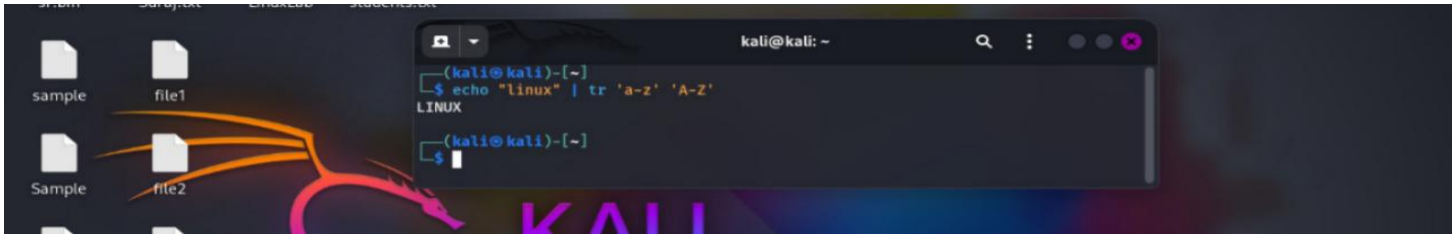
32 Extract the first column from a file using cut.



33 Compare two files using cmp.

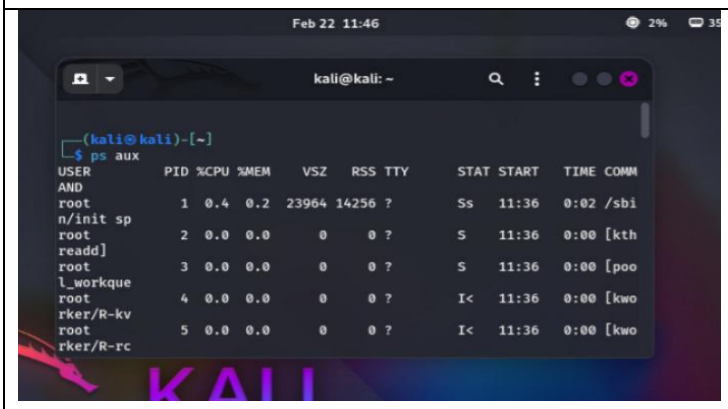


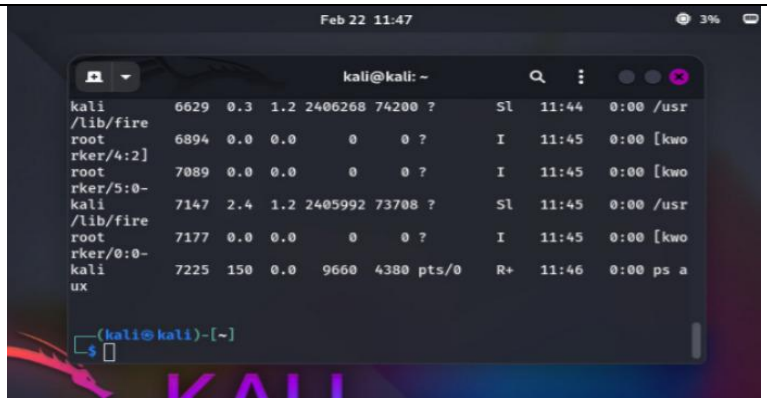
34 Convert lowercase letters to uppercase.



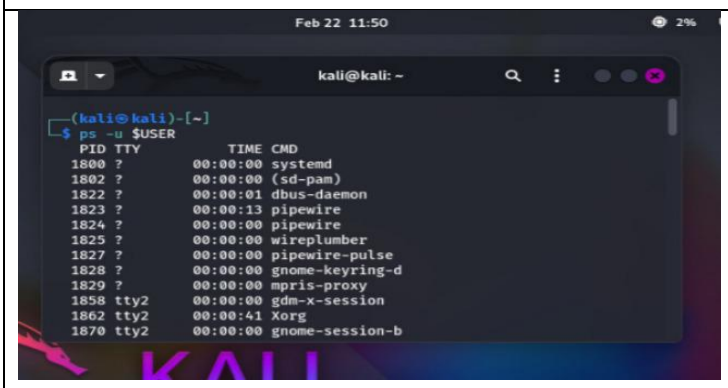
35 Save the output of a command into a file using tee.

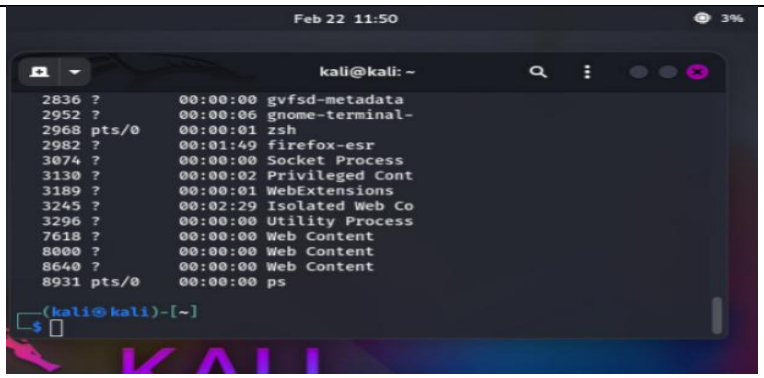
36 Display all running processes.



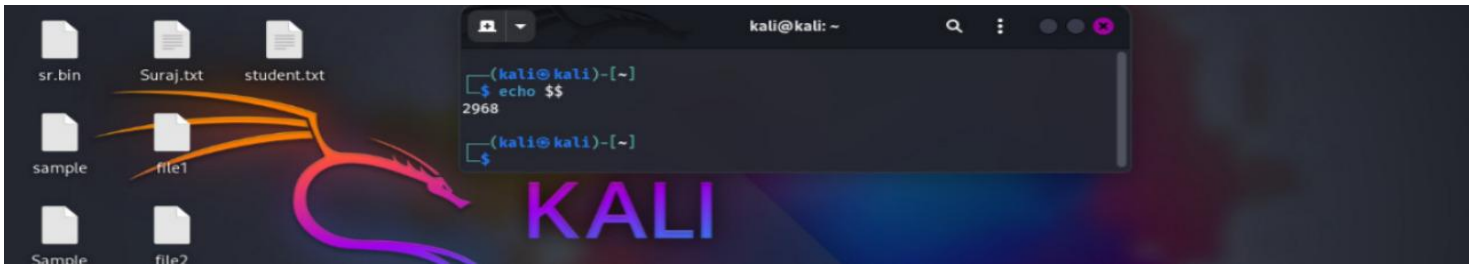


37 Display processes started by the current user.

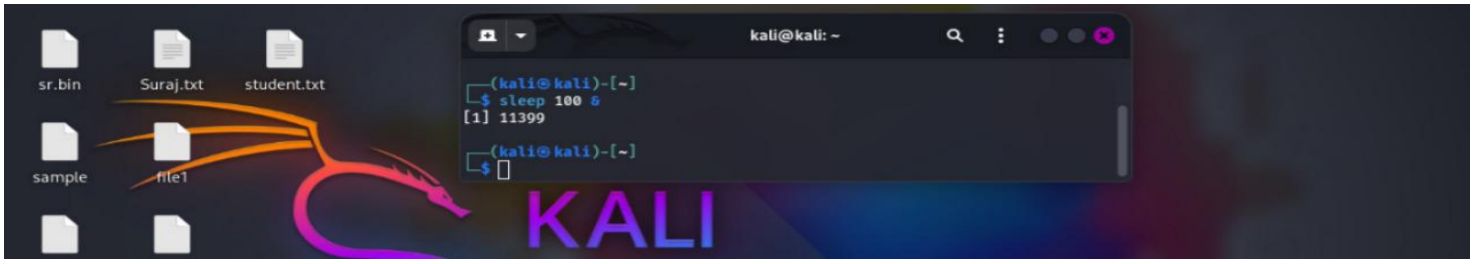




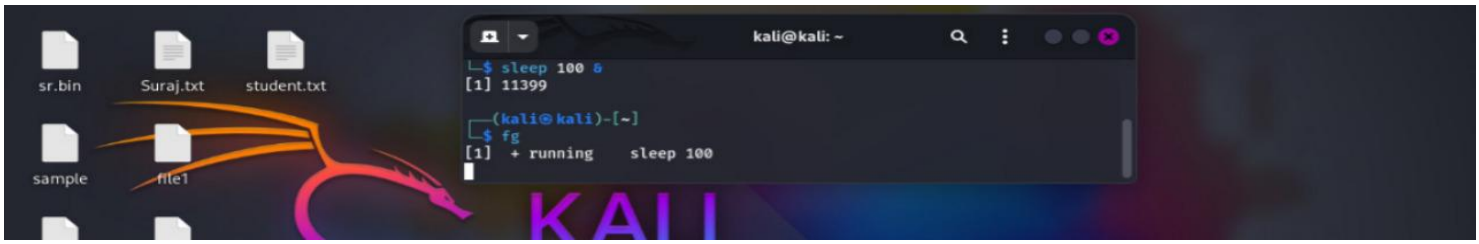
38 Find the process ID of the terminal.



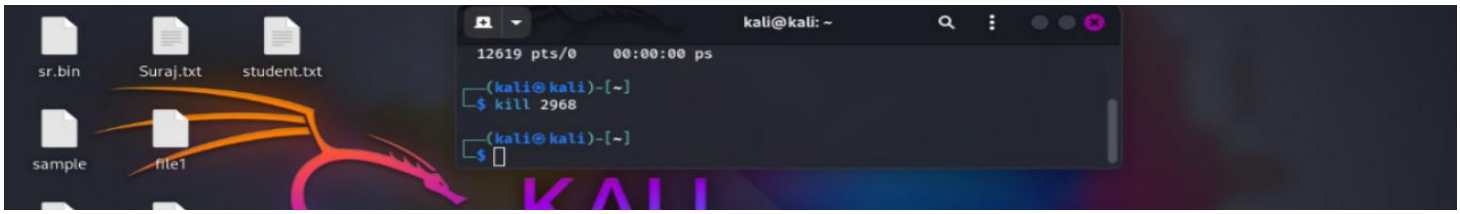
39 Run a command that sleeps for 100 seconds and send it to the background.



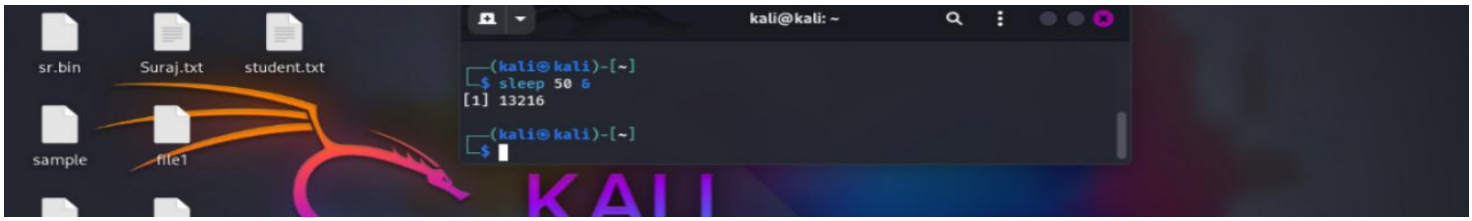
40 Bring a background job to the foreground.



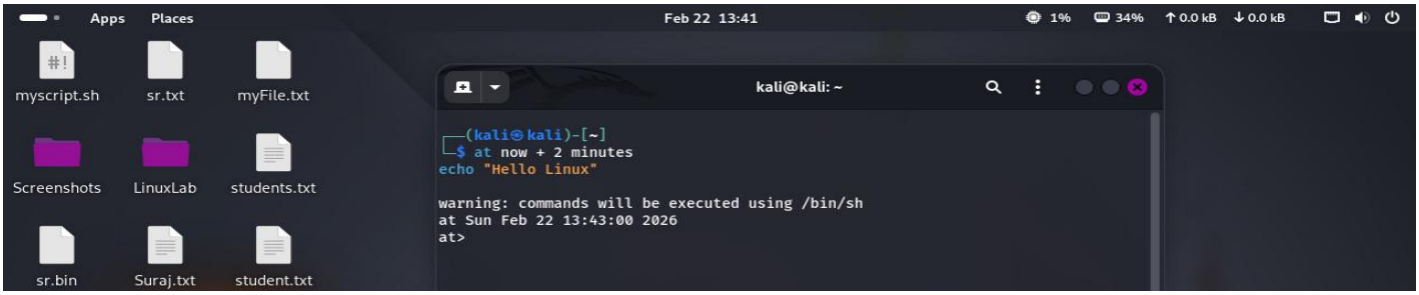
41 Kill a process using its PID.



42 Run a command directly in the background.

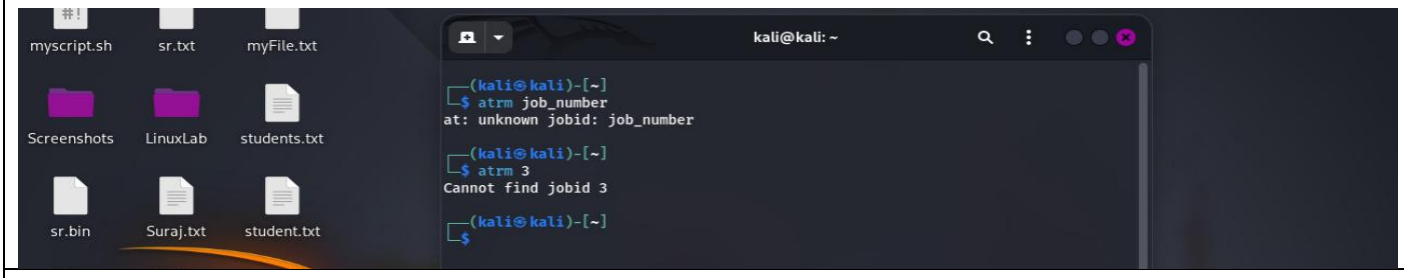


43 Schedule a job using at to display a message after 2 minutes.

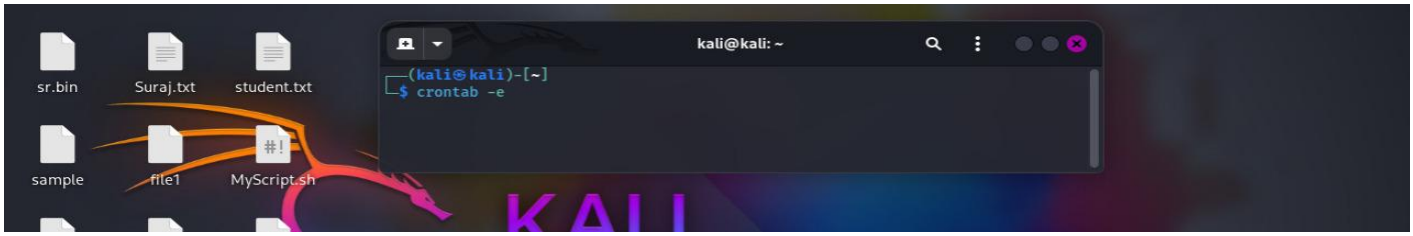


44 Display all scheduled jobs.

45 Remove a scheduled job.



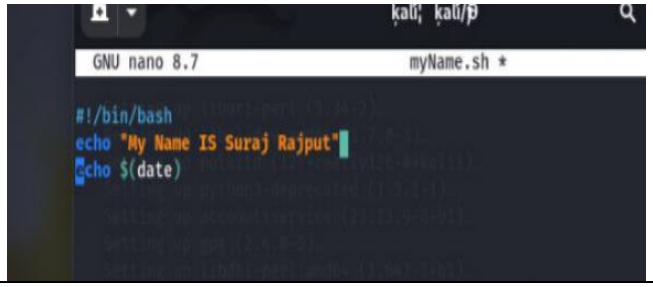
46 Create a cron job to display date and time every minute.

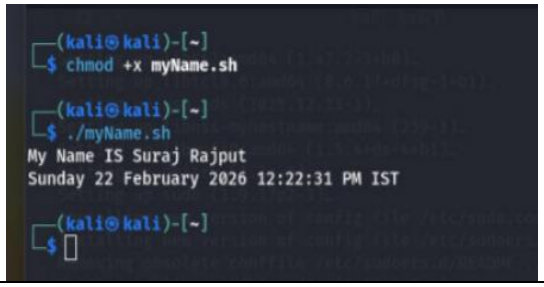


Unit – 3

1 Create a shell script to print "Welcome to Linux".

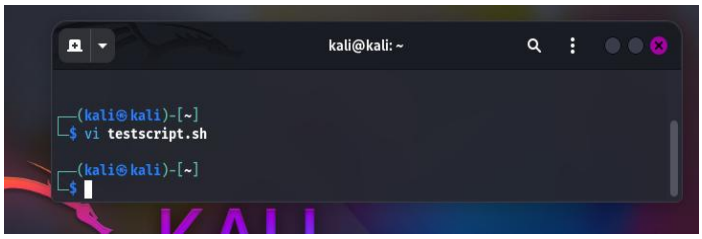
2 Create a script to display your name and current date.

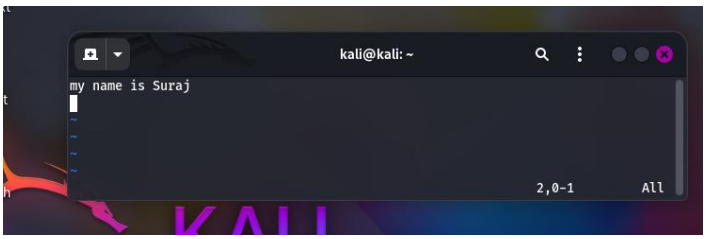




3 Create a script to accept a number and display it.

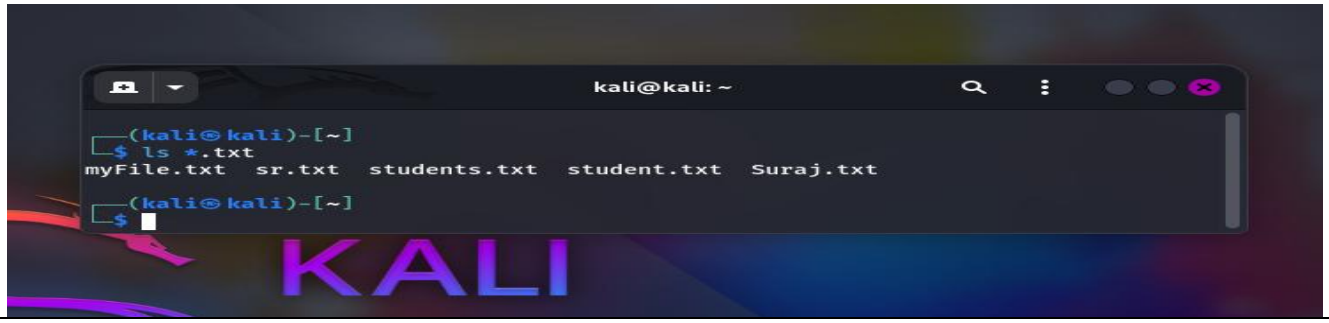
4 Create and execute a script using vi editor.



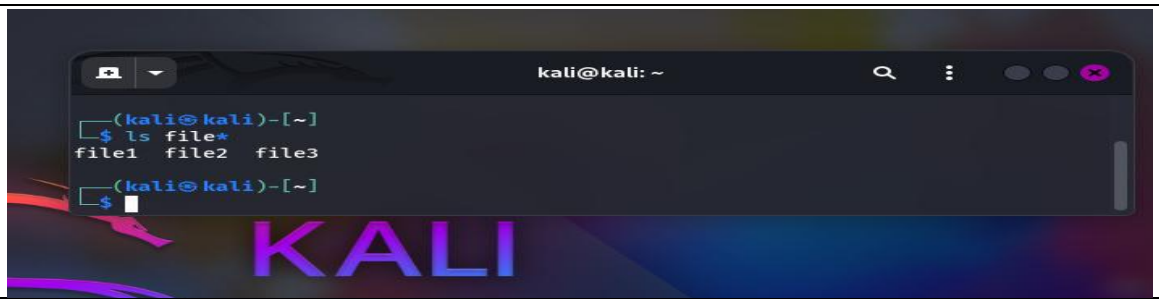


Pres Esc and Then :wq

5 Create files and list only .txt files using wildcards.

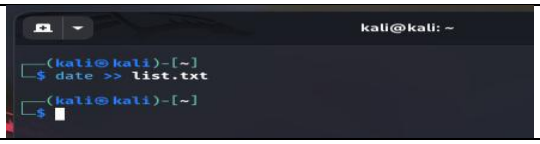


6 List all files starting with file.



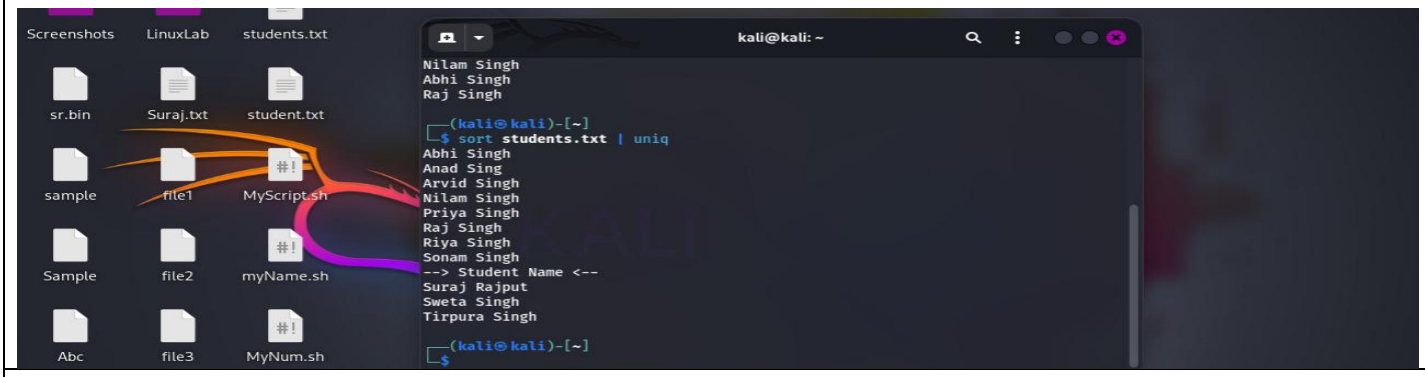
7 Redirect output of ls to a file.

8 Append current date to a file.

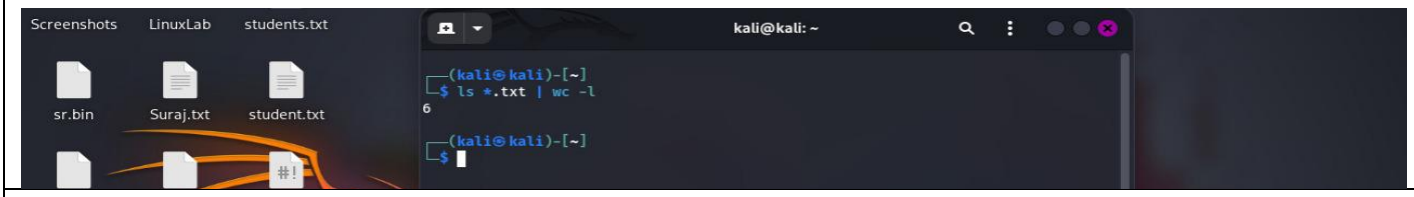


9 Read input from a file using redirection.

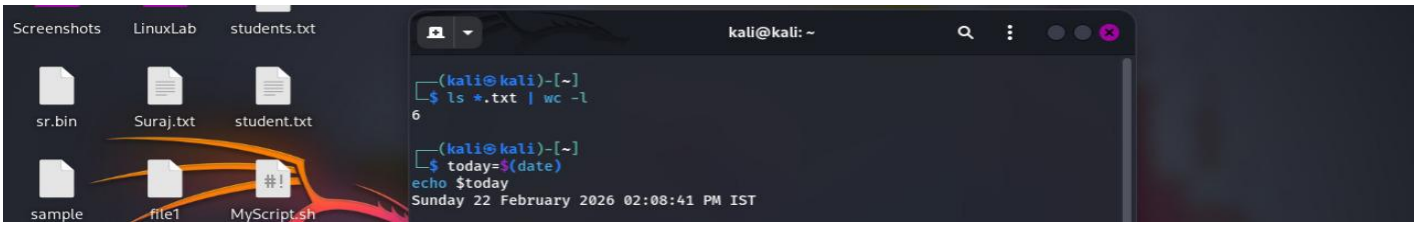
10 Display sorted unique names from a file using pipes.



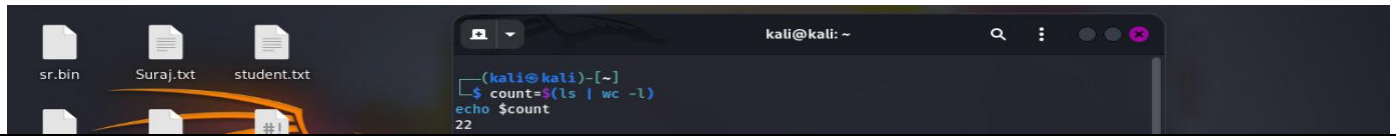
11 Count the number of .txt files using pipes.



12 Store today's date in a variable using command substitution.

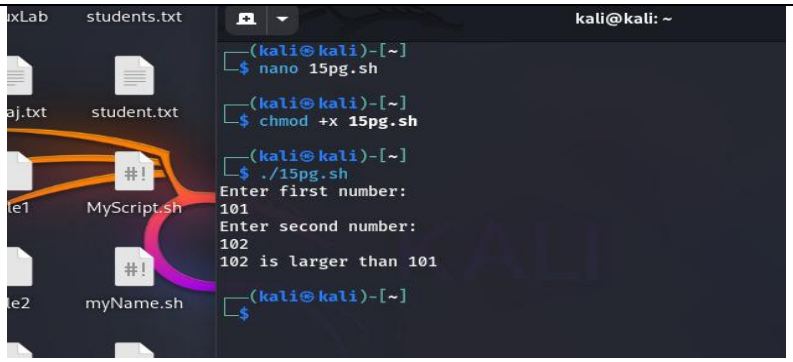


13 Store number of files in a directory into a variable

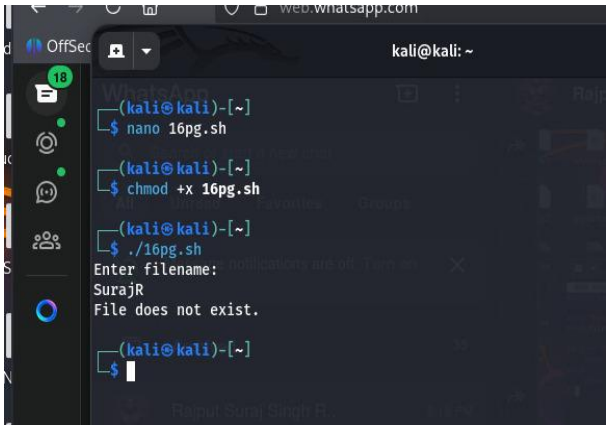


14 Write a script to check whether a number is even or odd.

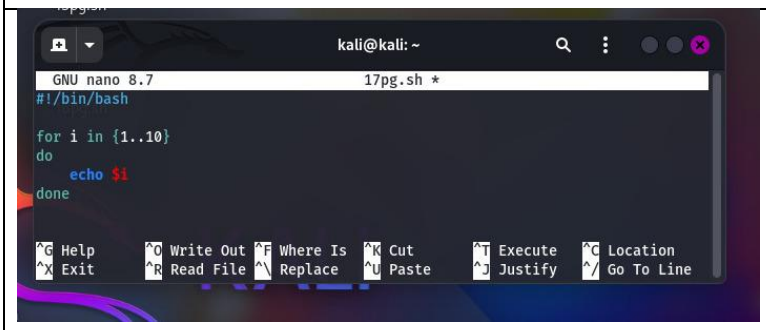
15 Write a script to find the largest of two numbers.



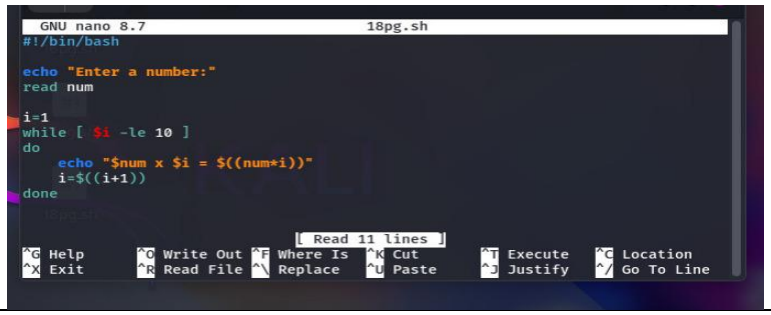
16 Write a script to check whether a file exists.



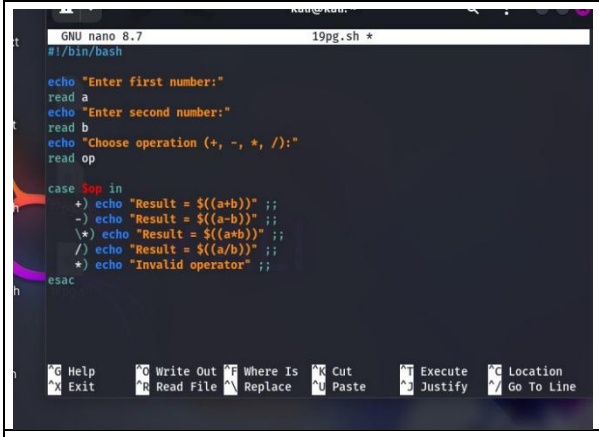
17 Write a script using for loop to print numbers from 1 to 10.

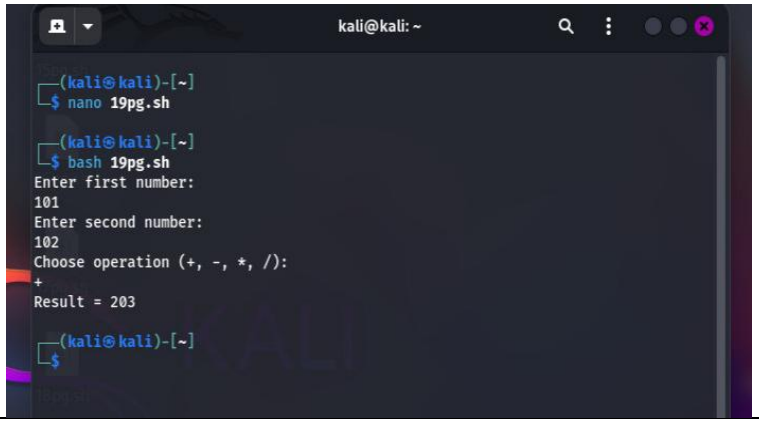


18 Write a script using while loop to print the table of a number.

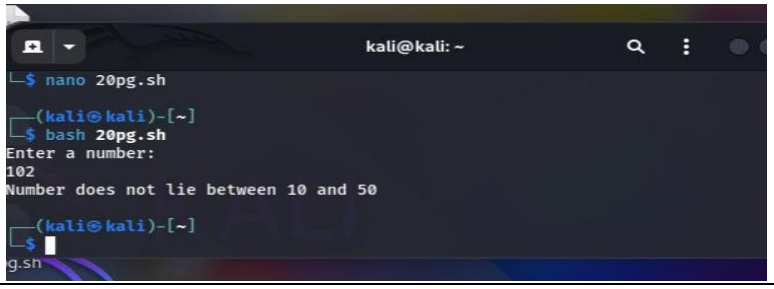


19 Write a script using case to implement a simple calculator.

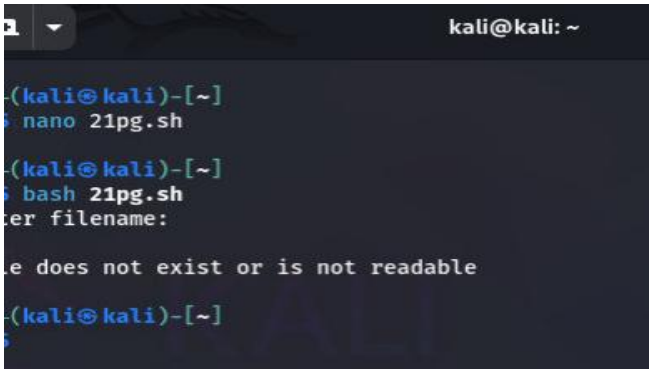




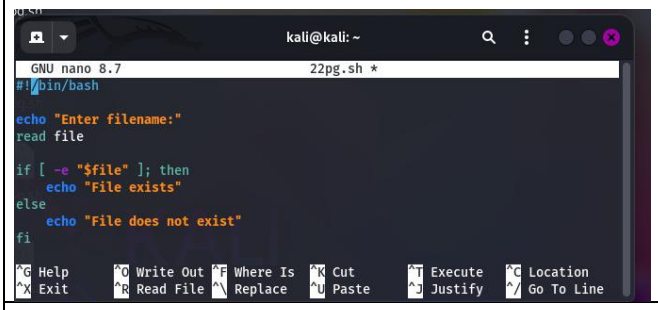
20 Write a script to check whether a number lies between 10 and 50

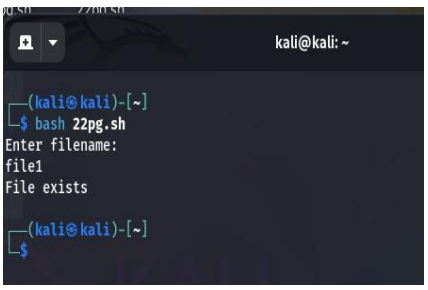


21 Write a script to check whether a file exists and is readable.

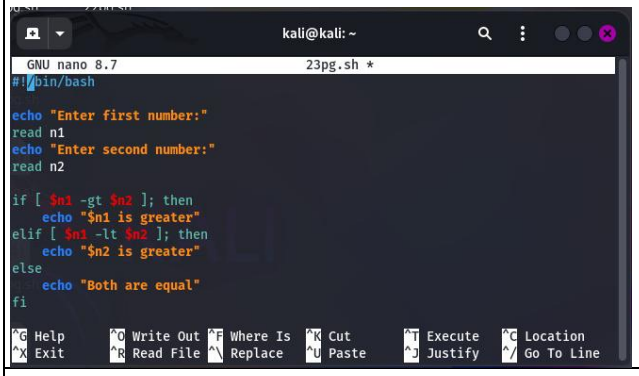


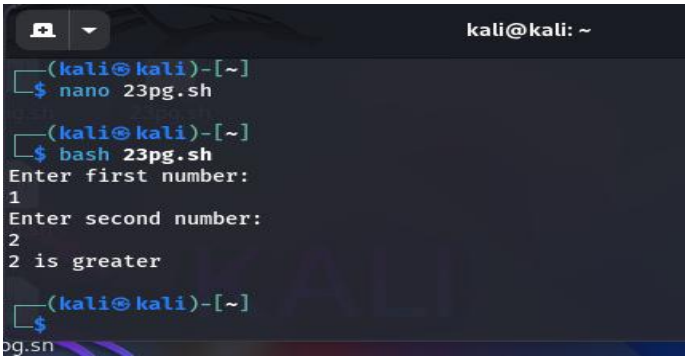
22 Write a script to check whether a file exists.



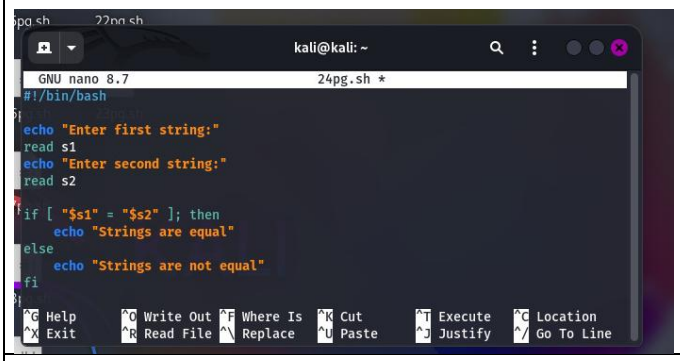


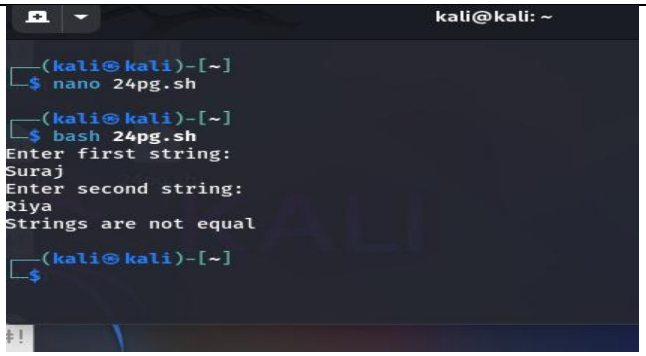
23 Write a script to compare two numbers.



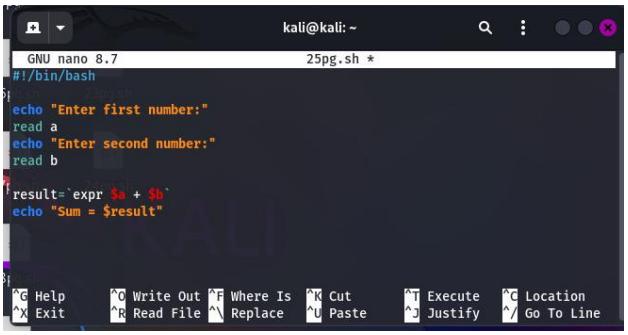


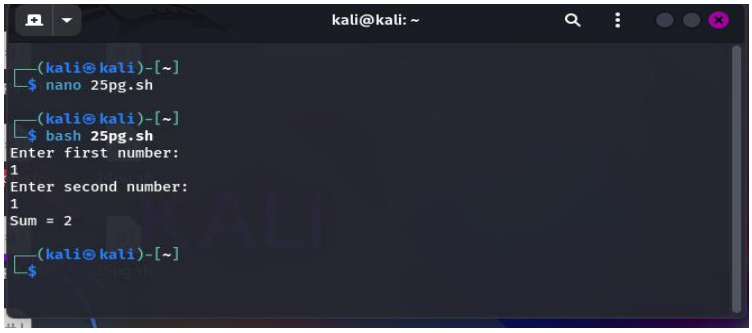
24 Write a script to compare two strings.



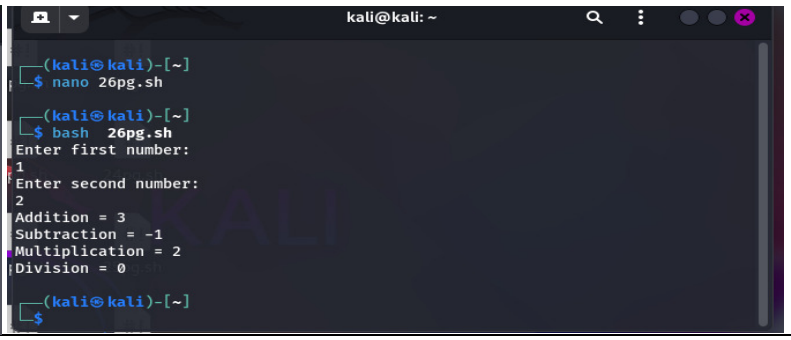


25 Write a script to add two numbers using expr.





26 Write a script to perform addition, subtraction, multiplication, and division using $(( )).



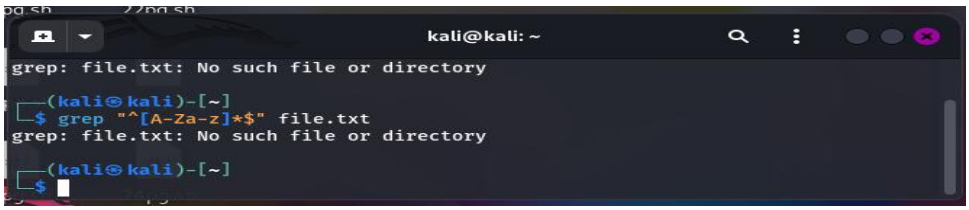
Unit -4

1 Find all lines starting with the word “Linux” from a file.

2 Find all lines ending with .txt.

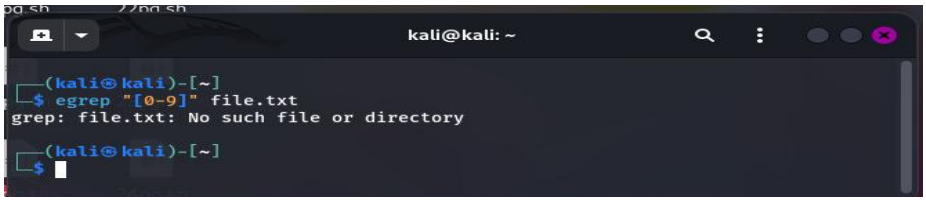
3 Find all lines containing exactly a 10-digit number.

4 Find all lines containing only alphabets.

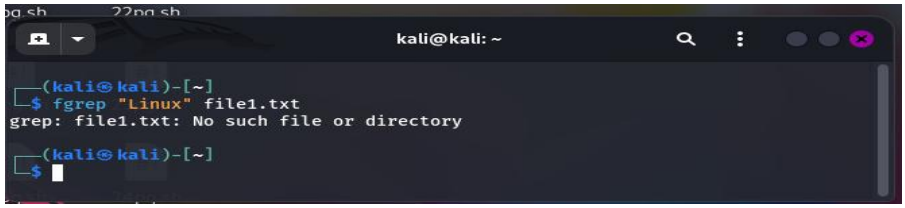


5 Search for the word “error” in a file.

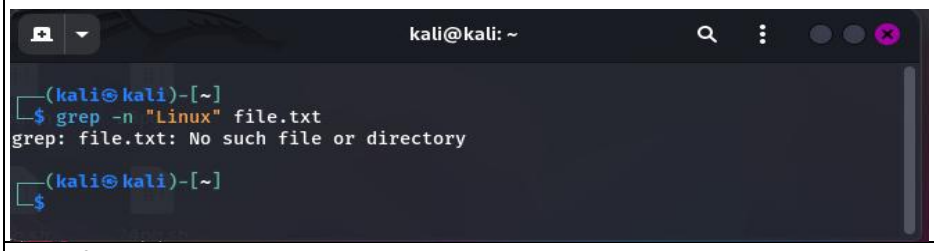
6 Search lines containing digits using egrep.



7 Search an exact string using fgrep.

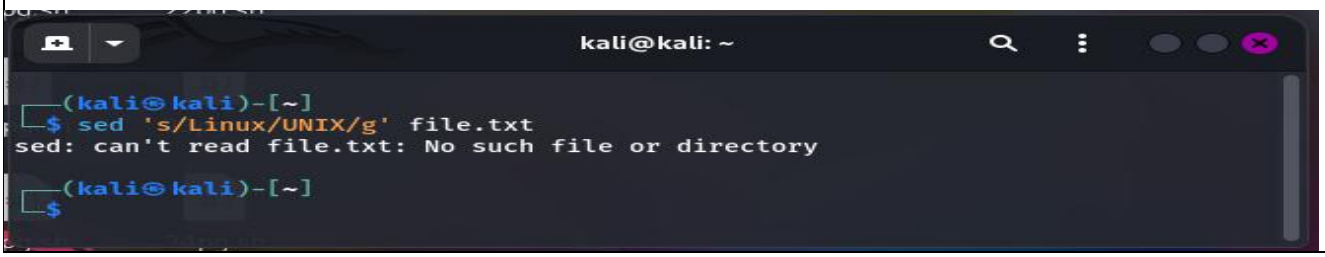


8 Display line numbers along with matched lines



9 Perform case-insensitive searching.

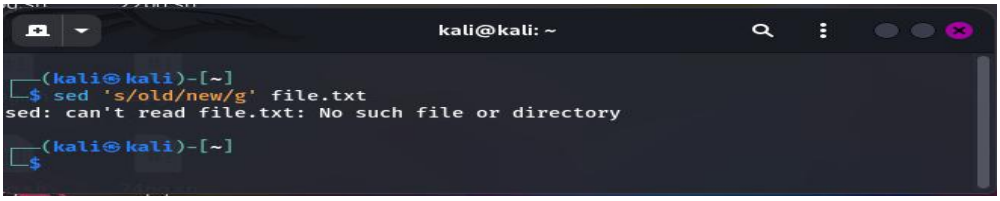
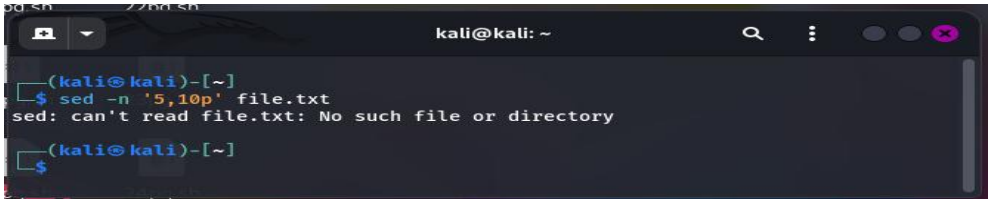
10 Replace the word “Linux” with “UNIX” in a file.



11 Delete the 3rd line from a file.

12 Insert a line at the beginning of a file.

13 Replace all occurrences of a word in a file.



14 Display lines 5 to 10 from a file.